

# Introduction to Formal Reasoning

## COMP2065-IFR

### L01 : Introduction

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# About Thorsten

## My Background

- Contact details:
  - Name: Thorsten Altenkirch
  - Office: C35
  - E-mail: Thorsten.Altenkirch@nottingham.ac.uk
  - Office hours : on demand (or just drop by)
- About me:
  - Graduated in 1989 at TU Berlin
  - Completed my PhD at U Edinburgh in 1994 (Type Theory)
  - Cochair of Functional Programming Laboratory
  - Interested in:
    - Functional Programming
    - Intuitionistic logic
    - Quantum Programming
    - Type Theory

# aboutlogic

Conversations on logic, mathematics, philosophy and computer science

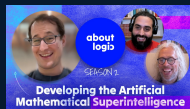
Hosted by **Thorsten Altenkirch** and **Deniz Sarikaya**



**Dana Scott**



**Emily Riehl**



**Tudor Achim**



**Neil Ghani**

Long-form conversations with leading researchers in the foundations of mathematics and computer science.

[youtube.com/@aboutlogic](https://youtube.com/@aboutlogic)

# Teaching Assistants

- Warren Jackson
- Ian Knight
- Christina O'Donnell
- Kunhong Du
- Samuel Toth
- Zhili Tian
- Mohammadaref Hajimohammadzadeh

# Contents

- **Formal logic and proofs**
- **Using Lean:** interactive proof system (we use **Lean 4**)
- **Topics**
  - Propositional logic
  - Classical logic
  - Predicate logic
  - Booleans
  - Natural numbers
  - Lists and trees

# What and when?

- See Moodle and your personal timetable for details and updates.
- **Lectures** (weeks 2–12; Business School South B52)
  - Monday 17:00–18:00
  - Friday 14:00–15:00
  - First lecture: Monday **28 September**
- **Computing labs** (weeks 3–12; A32 Computer Science)
  - Tuesday 11:00–12:00 **or** 12:00–13:00
  - First labs: Tuesday **6 October**
- **Tutorials** (weeks 3–12; one allocated slot, rooms vary)
  - Tuesday 16:00–17:00 or 17:00–18:00
  - Thursday 11:00–12:00
  - Friday 10:00–11:00 or 11:00–12:00
  - Start in the week beginning **5 October**

# Assessment

## What Should You Expect

- **Coursework and class test (50%)** **Important !!!**
  - Lean exercises** Frequent exercises, starting 5/10; usually weekly, with a deadline 10 days later.
  - Online quizzes** Short Moodle quizzes completed **in person in your allocated lab**. Together they count for **20% of the coursework mark**.
  - Class test** Under exam conditions (using Lean) on 8 December, in your lab slot: 11:00–12:00 or 12:00–13:00.
- **Exam (50%)**
  - Previous exams and a mock exam will be discussed in the revision lectures.

# Academic Misconduct

## The Riot Act

See the university policy

- Successful completion means that you are able to explain your solution during the lab.
- Do not post your solutions, coursework questions or exam questions anywhere.
- Do not seek assessment answers on social media or from other students.
- **AI can be useful as a tutor:** use it to ask for explanations, hints and feedback on your understanding.
- Do **not** use AI to do your coursework for you. You are responsible for understanding and being able to explain every solution you submit.

# Feedback

## What Should You Expect

- On Lean proof exercises: in lab and in tutorials
- Summary feedback on exam
- Individual exam feedback, and script viewing, on demand

# Enrolment and ECs

## Responsibilities of Student Services

- Enrolment, etc. is handled via student services.
- Late submissions and extensions for individual coursework will be handled by us. Please do not raise ECs.
- If you have continuous serious issues affecting the coursework or cannot attend the class test, please raise an EC (Extenuating circumstance) with student services.

## Contact & help

- **Forum (preferred)**: ask on Moodle first; others can help and answers are searchable.
- **Email (if personal)**: (include module code in the subject).
- **Office hours**: on demand (or just drop by).
- **Accessibility**: if you need adjustments, please contact me or Student Support early.

# Teaching Materials

- Main source: **moodle page**
- Material from lectures:
  - slides from lectures
  - proof examples
  - recording of interactive sessions
- Online lecture notes: linked from Moodle.

## Lean 4: install

- Install Visual Studio Code.
- In VS Code, install the official **Lean 4** extension (publisher: leanprover).
- The extension can guide you through installing **Elan**, the Lean version manager, and Lean itself.
- For the lectures and first exercises, **no Lake project is needed yet**.
- If you have installation problems, ask in the lab or post on the Moodle forum.

## Lean 4: module setup

- Create a local folder, e.g. `lean-lecture`. Avoid OneDrive/iCloud/Dropbox if possible, and open the **folder** in VS Code.
- We use **Lean 4.23.0**. In that folder create a file `lean-toolchain` containing:

```
leanprover/lean4:v4.23.0
```

- Create `Main.lean` and try:

```
def double (n : Nat) := n + n  
#eval double 3
```

You should see 6.

- If Lean gets stuck, use **Lean 4: Restart Server** from the Command Palette.

## Lean 4 in your browser

- For quick experiments you can use the official online editor:  
`live.lean-lang.org`.
- No installation is required: paste or type Lean code and wait for the Infoview to update.
- Quick test:

```
def double (n : Nat) := n + n  
#eval double 3
```

You should see 6.

- You can share an example by copying the URL for the current editor state.
- The web editor is excellent for small examples, but its Lean/Mathlib version may differ from the module setup. For assessed work and larger programs, use the local setup pinned to **v4.23.0**.

# AI and Formal Reasoning

- AI is an important **enabling technology for formal proof**: it can assist proof search and proof construction.
- It helps scale formalisation to **contemporary mathematics**— for example the *FLT project* (formalising Fermat's Last Theorem in Lean) and work on the *Navier–Stokes problem*.
- It can help us formally certify **realistic software and hardware systems**.
- For this module, use AI as a **tutor**: explanations, examples, hints and feedback. But **do not use it to do your coursework**—you need the practice.



*Do not take a forklift to the gym.*

## THREE LOGICIANS WALK INTO A BAR...

